

Measurement-Decoupled Attention–Koopman–T–S Fuzzy Monitoring via Structured Basis-Perturbation Residual Learning

Anonymous Authors

Abstract—Monitoring closed-loop nonlinear systems from normal data alone encounters a structural coupling: the measurements forming a residual may simultaneously adjust the encoder state, attention context, or fuzzy scheduling variable against which that residual is computed. We address this when physical fault channels, directions, temporal profiles, and labeled samples remain unavailable. A strictly causal attention encoder pairs with a Takagi–Sugeno fuzzy Koopman model to form a measurement-decoupled reference; new outputs evaluate residuals but never update the frozen rollout. A single start-indexed object unifies free-rollout training, short-window testing, and long-reference testing. The complete fuzzy Koopman Jacobian decomposes into interpolated local dynamics and membership-gradient contributions, yielding an infinity-norm bound implemented through elementwise hard parameterization—no eigenvalue, singular-value, inverse-matrix, or higher-derivative operation enters training. Structured basis perturbations inserted separately into input, measurement, and lifted coordinates produce finite-horizon residual responses that jointly reduce nominal propagation and increase group separability. Scores standardized by fuzzy region, command, command variation, and reference age admit calibrated dynamic thresholds. Sufficient residual-space conditions govern detection and unique-group decisions; when uniqueness fails, a compatible set replaces an unsupported singleton label.

Index Terms—Attention, fault detection, Koopman operator, normal-data-only learning, structured perturbation, Takagi–Sugeno fuzzy systems.

I. INTRODUCTION

Closed-loop nonlinear systems often span multiple operating regimes yet yield few labeled fault records. Constructing a monitor from normal trajectories alone is attractive but poses a fundamental difficulty: when an abnormal measurement enters an encoder, updates an attention context, or shifts a fuzzy scheduling variable, the resulting reference may track the fault trajectory rather than reveal the discrepancy required for detection.

Koopman operator theory pursues lifted coordinates that render nonlinear dynamics amenable to finite-dimensional linear approximation [1], [2]. Takagi–Sugeno (T–S) fuzzy models represent system behavior as normalized convex combinations of local linear descriptions [3]–[5]. Integrating these frameworks for modeling and control has been explored [6]; the integration itself is therefore not claimed as novel. Instead, we exploit the T–S structure in three distinct places: isolating the normal reference from measurement feedback, propagating finite-horizon errors through the complete fuzzy mapping, and conditioning

detection thresholds on the realized operating point. Attention mechanisms retain their capacity to summarize long input–output histories while being prevented from injecting new measurements into the frozen reference after its initialization.

Traditional fuzzy fault-detection approaches typically posit explicit fault or unknown-input channels, then synthesize observers or filters via Lyapunov arguments and linear matrix inequalities [7]–[11]. Koopman-based diagnosis has likewise leveraged learned residual generators and parity relations [12], [13]. Such designs presume access to a physical fault model or representative fault examples. Here, training data comprise exclusively normal segments; neither the physical location, direction, nor temporal signature of a fault is known. Consequently, no minimal detectable fault gain can be grounded in plant coordinates. The only tractable structure resides in the known input and output dimensions and in the coordinates of the learned lifted model.

This information constraint couples three design objectives. First, excluding new measurements from reference updates prevents contamination by abnormal data yet exposes the reference to uncorrected free-rollout drift. Second, reducing that drift alone is insufficient—a less sensitive model may shrink both normal and abnormal residuals. Third, an unknown physical fault direction precludes a justified low-dimensional projection onto a fault-sensitive subspace. The method must therefore bound normal propagation, preserve the joint residual in its native dimension, and learn distinguishable response patterns solely for explicitly defined model-coordinate perturbation groups.

The main contributions are threefold.

- 1) We construct a measurement-decoupled attention–Koopman–T–S reference paired with a start-indexed residual that unifies free-rollout training, short-horizon monitoring, and long-reference monitoring. An exact nonlinear residual recursion emerges, accompanied by a delay-confirmed anchor acceptance rule.
- 2) We decompose the complete T–S fuzzy Koopman Jacobian to expose contributions from local dynamics, rule overlap, local-model disparity, and scheduling sensitivity. Elementwise hard parameterization enforces an infinity-norm bound that yields finite-horizon normal residual bounds and reliable reference horizons without invoking spectral decompositions, singular-value computations, matrix inversions, or higher-order derivatives during training.

- 3) We integrate structured perturbation rollouts over input, measurement, and lifted coordinates with nominal propagation learning. The resulting input-conditioned statistic supports sufficient residual-space detection and group-separation conditions. When evidence for a unique group is lacking, the method returns a compatible set rather than an unjustified physical fault label.

Section II establishes the information boundary and formulates three aligned subproblems. Section III develops the model architecture, reference mechanism, propagation analysis, and structured response learning. Section IV presents the input-conditioned detection statistic and set-valued isolation logic. Section V concludes.

II. PRELIMINARIES AND PROBLEM FORMULATION

A. Information Boundary and Plant Description

Denote by $\mathbf{u}_k \in \mathbb{R}^{m_u}$ the controller command accessible to the monitor and by $\mathbf{u}_k^a$ the action delivered to the plant. The closed-loop plant evolves according to

$$\begin{aligned} \mathbf{u}_k^a &= \mathbf{u}_k + \mathbf{f}_k^a, \\ \mathbf{x}_{k+1} &= \mathcal{X}_0(\mathbf{x}_k, \mathbf{u}_k^a, \boldsymbol{\xi}_k) \mathbf{f}_k^p + \boldsymbol{\varepsilon}_k^x, \\ \mathbf{y}_k &= \mathcal{Y}_0(\mathbf{x}_k, \mathbf{u}_k^a, \boldsymbol{\xi}_k) \mathbf{f}_k^s + \boldsymbol{\varepsilon}_k^y, \end{aligned} \quad (1)$$

where $\mathbf{x}_k \in \mathbb{X} \subseteq \mathbb{R}^{m_x}$, $\mathbf{y}_k \in \mathbb{Y} \subseteq \mathbb{R}^{m_y}$, and $\boldsymbol{\xi}_k \in \mathbb{R}^{m_\xi}$ represent the physical state, measured output, and measured exogenous variable. The healthy mappings $\mathcal{X}_0$ and $\mathcal{Y}_0$ are unknown. Although the fault vectors $\mathbf{f}_k^a$, $\mathbf{f}_k^p$, and $\mathbf{f}_k^s$ indicate possible physical injection points, their dimensions provide no directional information to the learning algorithm.

The monitor observes $\mathbf{u}_k$, $\mathbf{y}_k$, and $\boldsymbol{\xi}_k$; the actual action $\mathbf{u}_k^a$ may be unavailable. Modeling the plant conditional on the recorded controller command answers: how would a healthy system respond under that command? When a feedback controller alters $\mathbf{u}_k$ after receiving an abnormal measurement, that altered command remains a known conditioning input, whereas any unmeasured discrepancy between $\mathbf{u}_k^a$ and $\mathbf{u}_k$ persists as a plant-side inconsistency.

Training and threshold calibration employ exclusively normal trajectories. No physical fault matrix, fault direction, fault derivative, fault temporal model, or labeled fault example is used. All model parameters and calibration quantities freeze during testing. We define the strictly past history

$$\mathcal{H}_k^- = \{\mathbf{u}_{k-\ell_h:k-1}, \mathbf{y}_{k-\ell_h:k-1}, \boldsymbol{\xi}_{k-\ell_h:k}\}, \quad (2)$$

which excludes the current output $\mathbf{y}_k$.

B. Finite-History Lifted Representation

Under healthy operation, Koopman evolution advances an observable through composition with the state transition [1]. An ideal finite vector of observables $\psi : \mathbb{X} \rightarrow \mathbb{Z} \subseteq \mathbb{R}^{m_z}$ would satisfy

$$\psi(\mathcal{X}_0(\mathbf{x}_k, \mathbf{u}_k, \boldsymbol{\xi}_k)) = \psi(\mathbf{x}_{k+1}). \quad (3)$$

We do not assume the existence of a finite invariant subspace or that the learned coordinates embed the physical state. Rather, a finite-history encoder estimates a predictive lifted

coordinate $\mathbf{z}_k^{\text{enc}} \in \mathbb{Z}$ from (2), and a T-S fuzzy interpolation of local lifted linear models approximates (3) over the normal operating domain. Residual propagation accounts for both approximation error and finite-history limitations, as derived in Section III.

C. Problem Formulation

Given the information boundary above, we address three subproblems.

- 1) *Reference construction*: Construct a normal reference that exploits new measurements for residual evaluation yet bars them from updating the reference itself, while employing a single residual definition across training and both short- and long-horizon testing.
- 2) *Normal propagation*: Quantify finite-horizon drift of that reference through the complete attention–Koopman–T–S composition using operations compatible with first-order network training, and render the realized input and fuzzy operating region explicit.
- 3) *Residual-space decision*: With normal data alone, jointly reduce normal residual propagation and enhance separability of reproducible model-coordinate perturbation responses; then construct an input-conditioned detection statistic that returns a unique group only when the residual pattern substantiates it.

III. PROPOSED METHOD

A. Attention–Koopman–T–S Normal Model

We embed each element of (2) as a token $\mathbf{h}_j$. For attention head $q \in \{1, \dots, m_{\text{hd}}\}$, the strictly causal coefficient assigning weight at sample k to past token j is

$$\varpi_{k,j}^{(q)} = \frac{\exp((\mathbf{W}_Q^{(q)} \mathbf{h}_{k-1})^\top (\mathbf{W}_K^{(q)} \mathbf{h}_j) / \sqrt{m_a})}{\sum_{\ell=k-\ell_h}^{k-1} \exp((\mathbf{W}_Q^{(q)} \mathbf{h}_{k-1})^\top (\mathbf{W}_K^{(q)} \mathbf{h}_\ell) / \sqrt{m_a})}. \quad (4)$$

Concatenating head outputs yields

$$\boldsymbol{\chi}_k = \mathbf{W}_O \left[\sum_{j=k-\ell_h}^{k-1} \varpi_{k,j}^{(q)} \mathbf{W}_V^{(q)} \mathbf{h}_j \right]_{q=1}^{m_{\text{hd}}}, \quad \mathbf{z}_k^{\text{enc}} = \mathcal{H}_\Theta(\mathcal{H}_k^-). \quad (5)$$

The encoder $\mathcal{H}_\Theta$ encompasses token embedding and multihead attention. Equation (4) renders the causal information flow explicit; attention itself is not treated as a separate contribution.

We form the scheduling vector as

$$\boldsymbol{\rho}_k = \mathbf{W}_z \mathbf{z}_k + \mathbf{W}_\chi \boldsymbol{\chi}_k + \mathbf{W}_u \mathbf{u}_k + \mathbf{W}_\xi \boldsymbol{\xi}_k. \quad (6)$$

Constant offsets appear via an augmented coordinate. For rule $i \in \{1, \dots, m_r\}$, define the negative metric energy and normalized firing strength by

$$\begin{aligned} \varphi_i(\boldsymbol{\rho}) &= -\frac{1}{2}(\boldsymbol{\rho} - \mathbf{c}_i)^\top \boldsymbol{\Pi}_i(\boldsymbol{\rho} - \mathbf{c}_i), \\ \omega_i(\boldsymbol{\rho}) &= \frac{\exp(\varphi_i(\boldsymbol{\rho}))}{\sum_{j=1}^{m_r} \exp(\varphi_j(\boldsymbol{\rho}))}. \end{aligned} \quad (7)$$

Hence $\omega_i \geq 0$ and $\sum_i \omega_i = 1$. The diagonal metric and center parameterizations read

$$\begin{aligned} \boldsymbol{\Pi}_i &= \text{Diag}[\underline{\pi} \mathbf{1} + (\bar{\pi} - \underline{\pi}) \text{sigmoid}(\tilde{\pi}_i)], \\ \mathbf{c}_i &= \bar{\mathbf{c}} \odot \tanh(\tilde{\mathbf{c}}_i). \end{aligned} \quad (8)$$

where operations inside the brackets are elementwise and $0 < \pi < \bar{\pi}$. Cross-coordinate coupling persists through (6); a diagonal metric in scheduling space does not impose axis-aligned partitions in the original history space.

The local and interpolated lifted models are

$$\begin{aligned} \mathbf{v}_i(\mathbf{z}, \mathbf{u}) &= \mathbf{A}_i \mathbf{z} + \mathbf{B}_i \mathbf{u}, \\ \mathcal{T}_\Theta(\mathbf{z}, \mathbf{u}, \boldsymbol{\xi}; \boldsymbol{\chi}) &= \sum_{i=1}^{m_r} \omega_i(\boldsymbol{\rho}) \mathbf{v}_i(\mathbf{z}, \mathbf{u}), \\ \mathbf{y}^{\text{dec}} &= \mathcal{D}_\Theta(\mathbf{z}, \mathbf{u}, \boldsymbol{\xi}). \end{aligned} \quad (9)$$

The decoder $\mathcal{D}_\Theta$ differs from the physical output map $\mathcal{Y}_0$ because the learned lifted state need not contain the physical state.

Propagation bounds stem from elementwise hard parameterizations applied to the transition matrices; for instance,

$$\begin{aligned} [\mathbf{A}_i]_{pq} &= \frac{\kappa_A}{m_z} \tanh([\tilde{\mathbf{A}}_i]_{pq}), \\ [\mathbf{B}_i]_{pq} &= \frac{\kappa_B}{m_u} \tanh([\tilde{\mathbf{B}}_i]_{pq}), \\ [\mathbf{W}_z]_{pq} &= \frac{\kappa_W}{m_z} \tanh([\tilde{\mathbf{W}}_z]_{pq}). \end{aligned} \quad (10)$$

Consequently $\|\mathbf{A}_i\|_\infty \leq \kappa_A$, $\|\mathbf{B}_i\|_\infty \leq \kappa_B$, and $\|\mathbf{W}_z\|_\infty \leq \kappa_W$ by construction, eliminating any need for spectral projections.

Training employs a normal split and a free-rollout horizon ℓ_{tr} . Initializing $\mathbf{z}_{s|s}^{\text{fr}} = \mathbf{z}_s^{\text{enc}}$, we propagate via predicted outputs rather than new measurements:

$$\begin{aligned} \boldsymbol{\chi}_{t|s}^{\text{fr}} &= \mathcal{H}_\Theta^\chi(\mathcal{H}_{t|s}^{\text{fr}}), \\ \mathbf{z}_{t+1|s}^{\text{fr}} &= \mathcal{T}_\Theta(\mathbf{z}_{t|s}^{\text{fr}}, \mathbf{u}_t, \boldsymbol{\xi}_t; \boldsymbol{\chi}_{t|s}^{\text{fr}}), \\ \mathbf{y}_{t|s}^{\text{fr}} &= \mathcal{D}_\Theta(\mathbf{z}_{t|s}^{\text{fr}}, \mathbf{u}_t, \boldsymbol{\xi}_t). \end{aligned} \quad (11)$$

The normal prediction loss aggregates discrepancies over rollouts:

$$\mathcal{L}_{\text{pred}} = \sum_s \sum_{\ell=1}^{\ell_{\text{tr}}} \left(\lambda_z \|\mathbf{z}_{s+\ell}^{\text{enc}} - \mathbf{z}_{s+\ell|s}^{\text{fr}}\|_1 + \lambda_y \|\mathbf{y}_{s+\ell} - \mathbf{y}_{s+\ell|s}^{\text{fr}}\|_1 \right). \quad (12)$$

Free rollout, in contrast to one-step teacher forcing, mirrors the online reference logic.

B. Measurement-Decoupled Reference and Unified Residual

For arbitrary start s , rollout (11) defines the lifted residual

$$\mathbf{e}_{k|s}^z = \mathbf{z}_k^{\text{enc}} - \mathbf{z}_{k|s}^{\text{fr}}, \quad k \geq s, \quad (13)$$

while output consistency yields

$$\mathbf{e}_k^y = \mathbf{y}_k - \mathcal{D}_\Theta(\mathbf{z}_k^{\text{enc}}, \mathbf{u}_k, \boldsymbol{\xi}_k). \quad (14)$$

Object (13) coincides with the training residual in (12). Online, choosing $s = k - \ell_{\text{sh}}$ produces a short rolling window. Let $\mathbf{s}_k^0$ denote the most recent delay-confirmed normal anchor; then

$$\mathbf{z}_k^0 = \mathbf{z}_{k|s_k^0}^{\text{fr}}, \quad \ell_k^0 = k - s_k^0. \quad (15)$$

Setting $s = s_k^0$ recovers the long-reference residual. We form the joint residual

$$\mathbf{e}_k = \begin{bmatrix} (\mathbf{e}_k^y)^\top & (\mathbf{e}_{k|k-\ell_{\text{sh}}}^z)^\top & (\mathbf{e}_{k|s_k^0}^z)^\top \end{bmatrix}^\top. \quad (16)$$

Training and testing thus share a unified lifted residual; only the rollout starting point varies.

Encoded states are not instantly committed as anchors. A candidate is stored and accepted after ℓ_{cf} samples pass without any warning in windows containing that candidate. During warnings and alarms, the current anchor freezes and (11) continues without measurement injection.

Proposition 1 (Delayed anchor acceptance). *Suppose every abnormal trajectory in a designated detectable class triggers a warning within ℓ_{det} samples of onset. If $\ell_{\text{cf}} \geq \ell_{\text{det}}$, then no candidate arising at or after onset is committed before the warning fires.*

Proof. Commitment requires ℓ_{cf} consecutive warning-free samples. Under the stated premise, a warning appears no later than $\ell_{\text{det}} \leq \ell_{\text{cf}}$ samples after onset, so any post-onset candidate violates the acceptance condition. $\square$

Proposition 1 is conditional; faults that leave the available input-output data unchanged cannot be detected. After an alarm clears, the method waits at least ℓ_h samples to ensure the encoder history no longer contains the flagged interval, then repeats delay confirmation.

Define the encoded-state innovation and context-mismatch terms by

$$\begin{aligned} \boldsymbol{\nu}_k &= \mathbf{z}_{k+1}^{\text{enc}} - \mathcal{T}_\Theta(\mathbf{z}_k^{\text{enc}}, \mathbf{u}_k, \boldsymbol{\xi}_k; \boldsymbol{\chi}_k), \\ \delta_{k|s}^\chi &= \mathcal{T}_\Theta(\mathbf{z}_k^{\text{enc}}, \mathbf{u}_k, \boldsymbol{\xi}_k; \boldsymbol{\chi}_k) \\ &\quad - \mathcal{T}_\Theta(\mathbf{z}_k^{\text{enc}}, \mathbf{u}_k, \boldsymbol{\xi}_k; \boldsymbol{\chi}_{k|s}^{\text{fr}}). \end{aligned} \quad (17)$$

Theorem 1 (Exact start-indexed residual propagation). *Assume $\mathcal{T}_\Theta$ is continuously differentiable in $\mathbf{z}$ along the segment joining $\mathbf{z}_{k|s}^{\text{fr}}$ and $\mathbf{z}_k^{\text{enc}}$. Then*

$$\mathbf{e}_{k+1|s}^z = \bar{\Gamma}_{k|s} \mathbf{e}_{k|s}^z + \boldsymbol{\nu}_k + \delta_{k|s}^\chi, \quad (18)$$

where

$$\bar{\Gamma}_{k|s} = \int_0^1 \Gamma_{z,k}(\mathbf{z}_{k|s}^{\text{fr}} + \tau \mathbf{e}_{k|s}^z) d\tau. \quad (19)$$

Consequently,

$$\mathbf{e}_{k|s}^z = \Phi(k, s) \mathbf{e}_{s|s}^z + \sum_{j=s}^{k-1} \Phi(k, j+1) (\boldsymbol{\nu}_j + \delta_{j|s}^\chi), \quad (20)$$

with $\Phi(k, j) = \bar{\Gamma}_{k-1|s} \cdots \bar{\Gamma}_{j|s}$ and $\Phi(j, j) = \mathbf{I}$.

The proof proceeds by adding and subtracting the transition evaluated at the encoded state and rollout context, then invoking the integral mean-value identity; details appear in Appendix A. Term $\boldsymbol{\nu}_k$ captures one-step normal-model mismatch, while $\delta_{k|s}^\chi$ retains the effect of finite attention memory.

C. Fuzzy-Structure-Dependent Normal Propagation

For brevity, write $\mathbf{v}_{i,k} = \mathbf{A}_i \mathbf{z}_k + \mathbf{B}_i \mathbf{u}_k$ and $\bar{\mathbf{v}}_k = \sum_i \omega_{i,k} \mathbf{v}_{i,k}$. From (7),

$$\begin{aligned} \gamma_{i,k}^\rho &:= \nabla_{\boldsymbol{\rho}} \varphi_{i,k} = -\boldsymbol{\Pi}_i(\boldsymbol{\rho}_k - \mathbf{c}_i), \\ \bar{\gamma}_k^\rho &:= \sum_j \omega_{j,k} \gamma_{j,k}^\rho, \\ \nabla_{\boldsymbol{\rho}} \omega_{i,k} &= \omega_{i,k} (\gamma_{i,k}^\rho - \bar{\gamma}_k^\rho). \end{aligned} \quad (21)$$

Theorem 2 (Complete fuzzy lifted-state Jacobian). *For fixed $\mathbf{u}_k$, ξ_k , and χ_k , the Jacobian of (9) with respect to $\mathbf{z}$ is*

$$\mathbf{\Gamma}_{z,k} = \sum_{i=1}^{m_r} \omega_{i,k} \mathbf{A}_i + \sum_{i=1}^{m_r} \omega_{i,k} (\mathbf{v}_{i,k} - \bar{\mathbf{v}}_k) (\gamma_{i,k}^\rho - \bar{\gamma}_k^\rho)^\top \mathbf{W}_z. \quad (22)$$

Moreover,

$$\begin{aligned} \|\mathbf{\Gamma}_{z,k}\|_\infty &\leq \kappa_k, \\ \kappa_k &:= \sum_i \omega_{i,k} \|\mathbf{A}_i\|_\infty \\ &\quad + \|\mathbf{W}_z\|_\infty \sum_i \omega_{i,k} \|\mathbf{v}_{i,k} - \bar{\mathbf{v}}_k\|_\infty \\ &\quad \cdot \|\gamma_{i,k}^\rho - \bar{\gamma}_k^\rho\|_1. \end{aligned} \quad (23)$$

Equation (22) separates the interpolated local dynamics from membership-gradient effects. The second term grows when rules overlap, their local predictions diverge, and the scheduling vector is sensitive in directions altering relative weights. This decomposition clarifies the fuzzy structure's specific role in reference analysis. Appendix B supplies the proof and induced-norm derivation.

Figure 1 depicts the overall information flow of the method, showing how strictly past history feeds the encoder, which produces the encoded state used both directly in residual formation and as input to the fuzzy Koopman model that generates the measurement-decoupled reference.

Assumption 1 (Normal residual forcing). *On a prescribed normal operating domain, known nonnegative scalars $\bar{\epsilon}_k$ satisfy*

$$\|\boldsymbol{\nu}_k + \boldsymbol{\delta}_{k|s}^\chi\|_\infty \leq \bar{\epsilon}_k \quad (24)$$

on the calibrated normal event.

The values in Assumption 1 are not claimed as deterministic global bounds obtained from finite data. The structural component emerges from (10)–(23); the unresolved remainder is calibrated on held-out normal blocks in Section IV.

Corollary 1 (Finite-horizon normal residual bound). *If $\beta_{s|s} \geq \|\mathbf{e}_{s|s}^z\|_\infty$ and*

$$\beta_{k+1|s} = \kappa_k \beta_{k|s} + \bar{\epsilon}_k, \quad (25)$$

then $\|\mathbf{e}_{k|s}^z\|_\infty \leq \beta_{k|s}$ while the trajectory remains within the prescribed normal domain. For an allowed reference tolerance $\bar{\beta}$, a reliable horizon is

$$\ell_{\text{ref}}(s) = \max\{\ell : \beta_{s+j|s} \leq \bar{\beta}, 1 \leq j \leq \ell\}. \quad (26)$$

All quantities in (23) involve matrix products, elementwise absolute values, and row or column sums. The empirical training counterpart of (25),

$$\mathcal{L}_{\text{nr}} = \sum_s \sum_{\ell=1}^{\ell_{\text{tr}}} \beta_{s+\ell|s}^{\text{emp}}, \quad (27)$$

serves solely as a normal-reference propagation regularizer. It does not, by itself, claim enhanced physical fault sensitivity.

D. Structured Basis-Perturbation Residual Learning

Reducing (27) addresses only the normal half of residual separation. To generate a diagnostic training signal absent physical fault examples, we define the model-coordinate group set

$$\mathbb{G}_b = \{u, y, z\}, \quad (28)$$

representing perturbations inserted into input, measurement, and lifted coordinates. These groups designate injection sites within the learned architecture; they are not physical actuator, sensor, or process labels.

At sampled start s , we select one standard basis vector and signed amplitude α . The input branch replaces $\mathbf{u}_s$ by $\mathbf{u}_s + \alpha \mathbf{q}_j^u$ in the model rollout. The measurement branch replaces $\mathbf{y}_s$ by $\mathbf{y}_s + \alpha \mathbf{q}_j^y$ in the current output residual and subsequent causal histories. The lifted branch employs

$$\mathbf{z}_{s+1}^{(z,j,\alpha)} = \mathcal{T}_\Theta(\mathbf{z}_s^{\text{enc}}, \mathbf{u}_s, \xi_s; \chi_s) + \alpha \mathbf{q}_j^z. \quad (29)$$

Thereafter, each branch follows the same frozen model and reference logic as the nominal branch.

Over horizon ℓ_b , we stack branch residuals and subtract the nominal stack:

$$\Delta \mathbf{e}_{s,\ell_b}^{g,j,\alpha} = \begin{bmatrix} (\mathbf{e}_s^{g,j,\alpha} - \mathbf{e}_s^0)^\top & \cdots & (\mathbf{e}_{s+\ell_b-1}^{g,j,\alpha} - \mathbf{e}_{s+\ell_b-1}^0)^\top \end{bmatrix}^\top. \quad (30)$$

The scaled response magnitude and unit- ℓ_1 pattern read

$$\eta_{g,j,\alpha}^{\text{rsp}} = \frac{\|\Delta \mathbf{e}_{s,\ell_b}^{g,j,\alpha}\|_1}{|\alpha| + \varepsilon_0}, \quad \mathbf{p}_{g,j,\alpha} = \frac{\Delta \mathbf{e}_{s,\ell_b}^{g,j,\alpha}}{\|\Delta \mathbf{e}_{s,\ell_b}^{g,j,\alpha}\|_1 + \varepsilon_0}, \quad (31)$$

where $\varepsilon_0 > 0$ provides numerical regularization.

Sensitivity and intergroup separation losses are

$$\begin{aligned} \mathcal{L}_{\text{sen}} &= \sum_{g,j,\alpha} [\bar{\eta}_g - \eta_{g,j,\alpha}^{\text{rsp}}]_+^2, \\ \mathcal{L}_{\text{iso}} &= \sum_{g \neq \bar{g}} [\eta_{\text{iso}} - \|\mathbf{p}_{g,j,\alpha} - \mathbf{p}_{\bar{g},j,\bar{\alpha}}\|_1]_+^2. \end{aligned} \quad (32)$$

The complete objective combines

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{nr}} \mathcal{L}_{\text{nr}} + \lambda_{\text{sen}} \mathcal{L}_{\text{sen}} + \lambda_{\text{iso}} \mathcal{L}_{\text{iso}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}}, \quad (33)$$

where $\mathcal{L}_{\text{bal}}$ penalizes rule occupancies falling below a prescribed minimum without enforcing uniform usage. Each minibatch samples one coordinate per group, so the additional computation amounts to three ordinary forward rollouts and first-order backpropagation. No Hessian, eigendecomposition, singular-value decomposition, matrix inverse, or online optimization appears.

IV. INPUT-CONDITIONED DETECTION AND SET-VALUED ISOLATION

A. Input-Conditioned Residual Standardization

The normal distribution of (16) depends on the realized command, command variation, fuzzy operating region, and reference age. Define the condition descriptor

$$\mathbf{v}_k = \mathbf{u}_k - \mathbf{u}_{k-1}, \quad \boldsymbol{\zeta}_k = [(\boldsymbol{\omega}_k^0)^\top \quad \mathbf{u}_k^\top \quad \mathbf{v}_k^\top \quad \xi_k^\top \quad \ell_k^0]^\top, \quad (34)$$

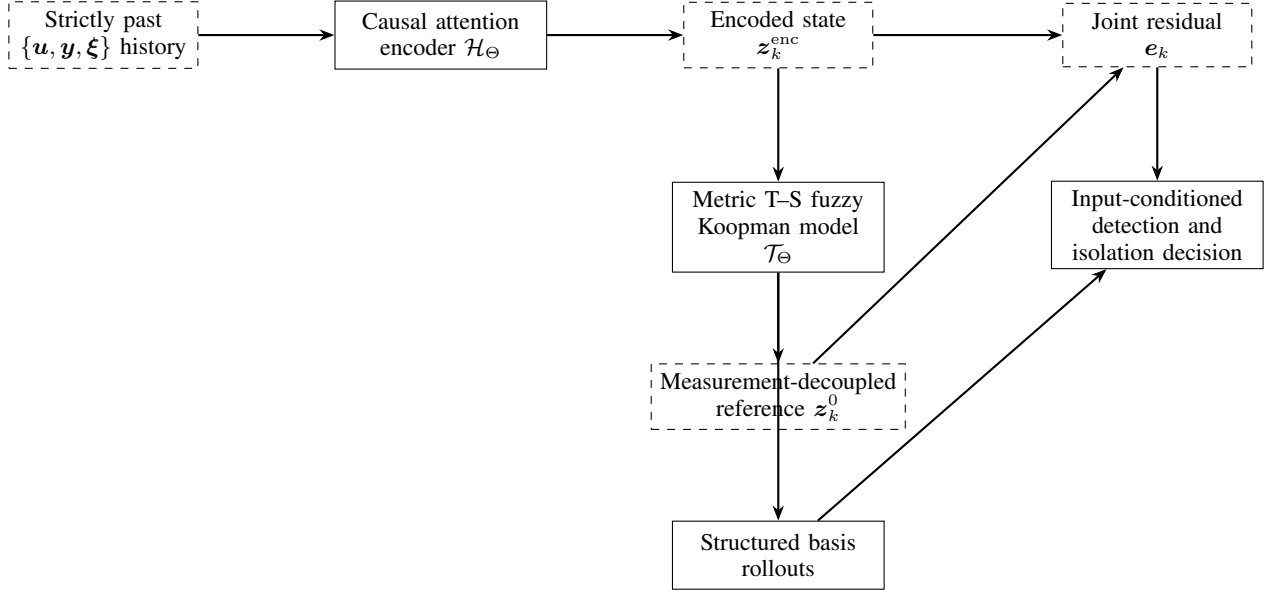


Fig. 1. Topology of the monitoring architecture. Strictly past history is processed by a causal attention encoder to produce the encoded state, which enters both the residual computation and the fuzzy Koopman model. The model generates a measurement-decoupled reference that contributes to the joint residual and feeds structured basis rollouts, while the decision stage integrates residual and rollout patterns under input-conditioned thresholds.

where ω_k^0 collects firing strengths evaluated along z_k^0 and its predicted context. For each rule, normal training blocks yield a local conditional mean $\mu_i(\zeta_k)$ and positive elementwise scale $\sigma_i(\zeta_k)$. Their mixture moments are

$$\begin{aligned}\mu_k &= \sum_i \omega_{i,k}^0 \mu_i, \\ \sigma_k^{\odot 2} &= \sum_i \omega_{i,k}^0 (\sigma_i^{\odot 2} + (\mu_i - \mu_k)^{\odot 2} + \varepsilon_\sigma \mathbf{1}).\end{aligned}\quad (35)$$

The between-rule term in (35) prevents underestimation of uncertainty at rule boundaries. Standardization proceeds elementwise:

$$\tilde{e}_k = (e_k - \mu_k) \oslash \sigma_k. \quad (36)$$

For each window length ℓ in a predeclared finite list, compute

$$s_{k,\ell} = \sum_{j=k-\ell+1}^k \|\tilde{e}_j\|_1. \quad (37)$$

A held-out normal calibration split is partitioned by predeclared fuzzy operating cells and reference-age intervals. Within each cell, nonoverlapping blocks calibrate the upper quantile $\tau_{\text{cal}}(\zeta_k, \ell)$. The propagated component from (25), after identical elementwise scaling, is denoted $\tau_{\text{prop}}(\beta_{k|s_k^0}, \ell)$. The online threshold becomes

$$\tau_{k,\ell} = \tau_{\text{cal}}(\zeta_k, \ell) + \tau_{\text{prop}}(\beta_{k|s_k^0}, \ell). \quad (38)$$

Thus the raw normal region adapts to input, input variation, fuzzy region, and reference age. The calibration term controls the unresolved stochastic remainder; the propagation term accounts for reference drift.

The multiscale statistic is

$$\varsigma_k^{\max} = \max_{\ell \in \{\ell_1, \dots, \ell_{\max}\}} \frac{s_{k,\ell}}{\tau_{k,\ell}}. \quad (39)$$

Its decision level is calibrated directly on the maximum score rather than applying separate uncorrected tests to each window. A finite-sample coverage interpretation requires exchangeable calibration and test blocks within each declared cell; under serial dependence, the guarantee is blockwise rather than samplewise [14], [15].

B. Sufficient Detection Condition

The following clarifies why normal propagation and structured response learning must operate jointly.

Proposition 2 (Residual-space detection). *For a selected window, suppose a normal standardized residual block $\tilde{e}^0$ satisfies $\|\tilde{e}^0\|_1 \leq \tau_{k,\ell}$ on the calibrated normal event. Let a structured perturbation yield the learned response $\Delta \tilde{e}^{g,j,\alpha}$, and suppose the mismatch between actual and learned responses is bounded by ε_g . If*

$$\|\Delta \tilde{e}^{g,j,\alpha}\|_1 > 2\tau_{k,\ell} + \varepsilon_g, \quad (40)$$

then the corresponding residual block exceeds $\tau_{k,\ell}$ on that event.

The proof invokes the reverse triangle inequality. If (32) ensures $\|\Delta \tilde{e}^{g,j,\alpha}\|_1 \geq \eta_g |\alpha|$, a sufficient amplitude condition is

$$|\alpha| > \frac{2\tau_{k,\ell} + \varepsilon_g}{\eta_g}. \quad (41)$$

Equation (41) is a residual-model conclusion, not a minimum physical fault claim. It identifies two controllable factors: reducing normal propagation lowers $\tau_{k,\ell}$, and amplifying the structured response raises η_g .

C. Unique-or-Set-Valued Isolation

Following detection, the observed residual increment is normalized as in (31). For group g , let $\delta_g(k)$ denote its smallest

ℓ_1 distance to response patterns stored from training rollouts. The compatible set is

$$\mathbb{G}_k^{\text{cmp}} = \{g \in \mathbb{G}_b : \delta_g(k) \leq \tau_{\text{iso}}\}. \quad (42)$$

A unique group is returned only when $\mathbb{G}_k^{\text{cmp}}$ contains one element and the distance to the second-best group exceeds a calibrated ambiguity margin.

Proposition 3 (Sufficient group separation). *Suppose the observed normalized response lies within δ_{obs} of a stored pattern from its generating group. If every pattern from a different group is at ℓ_1 distance exceeding $2\delta_{\text{obs}}$ from that pattern, nearest-pattern classification returns the generating group.*

The result follows from the triangle inequality. When the premise fails, (42) supplies the appropriate output. Labels u , y , and z indicate compatibility with an insertion location in the learned model. Absent an independently verified mapping, they must not be reinterpreted as unique actuator, sensor, or process faults.

D. Offline and Online Algorithms

Training, calibration, and testing employ disjoint normal trajectory blocks. Algorithm 1 summarizes the offline procedure.

Algorithm 1 Normal-Data Offline Design

Require: Disjoint normal training and calibration blocks

- 1: Train $\mathcal{H}_\Theta$, $\mathcal{T}_\Theta$, and $\mathcal{D}_\Theta$ by free rollouts using (12).
 - 2: Apply (8) and (10); compute (23) and (27).
 - 3: Sample one coordinate from each group in (28); optimize (33) with ordinary forward branches.
 - 4: Freeze all model and fuzzy-rule parameters.
 - 5: Estimate (35) and calibrate (38) on held-out nonoverlapping normal blocks.
 - 6: Store response patterns and calibrate the ambiguity margin for (42).
-

Online, the encoder continuously reads the strictly past history, while the normal reference either advances from its accepted anchor or awaits delayed reconfirmation. Algorithm 2 outlines the decision loop.

Algorithm 2 Online Monitoring

Require: Frozen model, normal statistics, thresholds, and response patterns

- 1: **for** each sample k **do**
 - 2: Compute z_k^{enc} and e_k^y from the data branch.
 - 3: Propagate the short rollout and the measurement-decoupled reference.
 - 4: Form e_k by (16) and ζ_k by (34).
 - 5: Compute (36)–(39).
 - 6: **if** ζ_k^{max} exceeds its calibrated decision level **then**
 - 7: Freeze anchor acceptance and report an alarm.
 - 8: Return $\mathbb{G}_k^{\text{cmp}}$ by (42).
 - 9: **else**
 - 10: Continue delay-confirmed anchor management.
 - 11: **end if**
 - 12: After an alarm, wait ℓ_h samples and repeat delay confirmation before accepting a new anchor.
 - 13: **end for**
-

The online stage executes encoder and model forward passes, elementwise standardization, vector norms, and table or lightweight-head lookups. It updates neither fuzzy centers, metrics, network weights, nor thresholds.

V. CONCLUSION

This paper has formulated normal-data-only nonlinear monitoring as three interconnected subproblems: barring new measurements from updating a normal reference, bounding the ensuing free-rollout drift, and making residual-space decisions absent physical fault priors. A measurement-decoupled attention–Koopman–T–S reference paired with a start-indexed residual resolved the training–testing discrepancy. Decomposing the complete fuzzy Jacobian linked normal propagation to local dynamics, rule overlap, local-model disparity, and scheduling sensitivity via an implementable infinity-norm bound. Structured model-coordinate perturbations complemented the normal propagation objective by amplifying residual responses and intergroup separation. Input-, fuzzy-region-, and reference-age-conditioned standardization produced a dynamic decision rule equipped with residual-space detection and set-valued isolation conditions. The method does not claim unique physical fault identification; such a claim requires additional actuation measurements, fault-channel knowledge, or a verified mapping from lifted coordinates to physical components.

APPENDIX A PROOF OF THEOREM 1

From (13), add and subtract $\mathcal{T}_\Theta(z_k^{\text{enc}}, u_k, \xi_k; \chi_{k|s}^{\text{fr}})$ and then $\mathcal{T}_\Theta(z_k^{\text{enc}}, u_k, \xi_k; \chi_k)$. The final difference equals ν_k , and the context difference is $\delta_{k|s}^\chi$. The remaining state difference equals (19) $e_{k|s}^z$ by the integral mean-value identity. This establishes (18). Repeated substitution produces (20).

APPENDIX B

PROOF OF THEOREM 2

Differentiate $\mathcal{T}_\Theta = \sum_i \omega_i \mathbf{v}_i$. Since $\partial \mathbf{v}_i / \partial \mathbf{z} = \mathbf{A}_i$ and $\partial \boldsymbol{\rho} / \partial \mathbf{z} = \mathbf{W}_z$,

$$\boldsymbol{\Gamma}_{z,k} = \sum_i \omega_{i,k} \mathbf{A}_i + \sum_i \mathbf{v}_{i,k} (\nabla_{\boldsymbol{\rho}} \omega_{i,k})^\top \mathbf{W}_z. \quad (43)$$

Substituting (21) and noting $\sum_i \omega_i (\nabla \varphi_i - \sum_j \omega_j \nabla \varphi_j) = 0$ permits replacing $\mathbf{v}_{i,k}$ by $\mathbf{v}_{i,k} - \bar{\mathbf{v}}_k$, yielding (22).

For vectors $\mathbf{a}$ and $\mathbf{b}$, $\|\mathbf{a}\mathbf{b}^\top\|_\infty = \|\mathbf{a}\|_\infty \|\mathbf{b}\|_1$. The triangle inequality and submultiplicativity then deliver (23).

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